Response to Anonymous Referee #2

Manuscript Title: Climatology and trends of extreme precipitation in France: evaluation of an explicit-convection regional climate model

Authors: Nicolas Decoopman, Juliette Blanchet, Antoine Blanc, and Cecile Caillaud

Journal: Hydrology and Earth System Sciences (HESS)

We thank referee #2 for their constructive and detailed review of our manuscript. Below are our point-by-point responses to the comments, with line and figure numbers referring to the manuscript.

General Comments

Comment 1: Return Period Choice

There is a 10-year return period chosen for both daily and hourly data, even though 10-year return period for hourly is much more extreme compared to daily given the amount of hourly data in a 10-year period compared to daily. Has a lower return value been tested? Given the noisy result, and the shorter period of data a better justification for the high return period for hourly data should be given, and it is of interest to show a lower return period.

- **Response:** We thank the reviewer for this comment. We agree that estimating a 10-year return period (RL_{10y}) at the hourly scale from a 33-year series (1990-2022) introduces significant sampling noise. To address this, we have also analyzed lower return periods, specifically 2-year and 5-year return levels (RL_{2y} and RL_{5y}), which are closer to the bulk of the annual maxima distribution and thus more robustly estimated.

Our tests indicate that while the spatial patterns of the trends are smoother and more robust, the main conclusions regarding the model's limitations in capturing hourly trends remain unchanged. We have added a stronger justification for the choice of the 10-year return period (as it is a standard design threshold in hydrology) while discussing the benefits of looking at lower return levels and presenting their trend maps in the supplementary material (Appendix B).

Manuscript modifications:

- In `Section 3.2.3 (Trend in return level)` (formerly line 377), we have added the justification for using the 10-year return level as the main indicator, while explaining the interest in comparing it with lower return levels:

  > *"In this study, we focus on the 10-year return level (*RL_{10}*) as the primary metric for extreme precipitation, as it represents a standard design threshold in hydrology. However, estimating a 10-year return level at the hourly scale from a relatively short record (33 years) is subject to significant sampling uncertainty. To assess the sensitivity and robustness of the detected trends, we also perform a sensitivity analysis using lower return periods (2-year and 5-year return levels,* RL_2 *and* RL_5*), which are closer to the center of the annual maxima distribution and thus more robustly estimated (see Appendix B)."*

- We have created a new appendix, `Appendix B: Sensitivity of GEV trends to the return period choice`, showing the trend maps for the 2-year and 5-year return levels (Figure B1 for daily, Figure B2 for hourly).

Comment 2: Introduction & Physical Processes

Introduction: First paragraph introduces a lot of different processes that are not necessarily connected. For instance, global warming leads to an increase in surface temperature, but it's the temperature increase in the atmosphere that increases the water holding capabilities, and the energy balance is also between the top of atmosphere and surface (and everything in between). A lot of the statements in Introduction need a reference for instance line 47-50 and should include more research articles in addition on the IPCC 2021 report.

- **Response:** We agree that the physical description in the first paragraph was somewhat compressed and mixed different thermodynamic and dynamic concepts. In the revised manuscript, we have restructured this paragraph to:

1. Clarify the physical link: surface warming leads to atmospheric warming, which increases the water-holding capacity of the troposphere according to the Clausius-Clapeyron relationship, while the actual condensation and precipitation are driven by vertical motion and adiabatic cooling of ascending air masses.
2. Add more academic references (e.g., Trenberth et al. 2003, O'Gorman and Muller 2010, Westra et al. 2014) alongside the IPCC 2021 report to ground these statements in the literature.

Manuscript modifications:

- In `Section 1 (Introduction and context)`, the first paragraph has been rewritten and expanded to clarify the physical links and integrate the suggested references:

"Global warming has reached +1.1°C worldwide, +1.7°C in metropolitan France, and +2°C in the French Alps compared to the pre-industrial era. As the troposphere warms, its capacity to hold moisture increases by approximately 7% per °C of warming, according to the Clausius-Clapeyron relationship (Clapeyron, 1834; Trenberth et al., 2003; Westra et al., 2013). However, the response of precipitation to warming is highly dependent on the event intensity. For non-extreme or average precipitation events, energetic constraints (such as global surface evaporation and radiative cooling) and dynamic constraints (such as subsidence and synoptic-scale atmospheric circulation changes) limit the increase in mean precipitation to only 1–3% per °C (O'Gorman and Muller, 2010; IPCC, 2021). In contrast, during intense convective events, such as local thunderstorms, the rapid updraft of warm, moist air condenses nearly all of this surplus water vapor. As a result, short-duration extreme precipitation increases at a rate close to the Clausius-Clapeyron scaling of 7% per °C—and can locally exceed it due to convective feedbacks—whereas mean rainfall remains tightly constrained by global energy budgets. Thus, climate warming leads to a robust increase in extreme precipitation, though this trend varies with regional atmospheric dynamics (O'Gorman and Muller, 2015; Blanchet et al., 2021)."

Comment 3: Treatment of Aerosols and Solar Brightening

Figure 2 and Figure 3: How much of the temperature trend could be caused by aerosol changes due to anthropogenic pollution? How is aerosol treated in AROME? The results in these plots indicate that there is an increase in temperature after the peak pollution in the 1980s and that the trend in 10-year return level is increasing after this peak. Even though aerosols are not included or kept constant in AROME, this should be stated and included in the discussion and limitations.

- **Response:** We would like to clarify that aerosols are **not** kept constant in this AROME simulation: they are monthly evolving and vary from year to year, taken from the ALDERA reanalysis (which uses an interactive aerosol scheme to capture historical variations, including the peak pollution in the 1980s and subsequent solar dimming/brightening, as described by Nabat et al. 2020). Green House Gases also evolve annually. In the revised manuscript, we have made this clearer in Section 2.2 and Section 5.1.1 (Discussion).

Manuscript modifications:

- In `Section 2.2 (CNRM-AROME model)`, we have clarified the description of how aerosols and greenhouse gases are treated in the simulation:

  "Aerosols are not kept constant in this simulation; they are monthly evolving and vary from year to year, taken from the ALDERA reanalysis (which uses an

interactive aerosol scheme to capture historical variations, including the peak anthropogenic pollution in the 1980s and subsequent solar dimming/brightening, as described by Nabat et al., 2020). Greenhouse gases also evolve annually."

- In `Section 5.1.1` (Discussion, Model limitations), we have added a discussion of the temperature trend underestimation and the role of aerosols:

  "It is also worth discussing the role of aerosols and solar dimming/brightening in these temperature trends. In this AROME simulation, aerosols are not kept constant: they are monthly evolving and vary from year to year, taken from the ALDERA reanalysis (which uses an interactive aerosol scheme to capture historical variations, including the peak anthropogenic pollution in the 1980s and subsequent solar dimming/brightening, as described by Nabat et al., 2020). Greenhouse gases also evolve annually. Thus, AROME includes the radiative forcing changes associated with these historical variations, but the weaker simulated temperature trend suggests that either the model's sensitivity to these forcings or the boundary conditions provided by ERA5 lead to a dampened warming signal in France compared to observations."

Comment 4: Visualization of Non-Significant Trends

For the maps with stations, setting non-significant trends to zeros makes it difficult to differentiate with stations that have close to zero or zero significant trends and stations with strong trends that are non-significant. The authors should consider not showing the stations at all or choosing another marker. The size of the marker also seems to vary with the trend, which can make it look like the stations with strong trends are more numerous than they are.

- **Response:** We agree with the reviewer. Representing non-significant trends as zero makes it difficult to distinguish between a significant trend of 0% and a non-significant trend of 50%. To improve readability:

1. We represent non-significant stations using a small, neutral, light-grey dot to keep them visible as part of the network but clearly distinguish them from stations with significant trends.
2. As also requested by Reviewer 1, we keep the marker size constant for all stations to prevent visual bias towards larger circles, relying solely on color to represent the trend magnitude.

Manuscript modifications:

- We have updated all maps showing station-based trends (Figures 6, 7 and 8) to represent stations with non-significant GEV trends using a small, neutral light-grey dot, while stations with significant trends are represented with a uniform-sized marker colored according to the trend value.

- The caption of Figure 6 (formerly Figure 5) has been updated to reflect these visual changes:

> *"Figure 6. Relative GEV trends over 1959–2022 in the 10-year return level of daily precipitation for AROME (left) and Météo-France stations (right) with the spatial correlation (r), the number of stations compared (n), and the mean error (ME). Only statistically significant trends (at the 10% level) are colored; stations with non-significant trends are represented by small light-grey dots."*

- Similar updates have been made to the captions of Figure 7 (formerly Figure 6) and the monthly Refinement Figure (Appendix C, Figure C1).

Minor Comments

- **Why choose the hydrological year? The study itself does not take into account snowpack, and in addition the convective extreme precipitation can come in summer that in this study is over the start of the hydrological year (JAS, where summer usually is considered as JJA). This should be justified more.**

  Response: The choice of the hydrological year (September 1 to August 31) is standard in French climatology and hydrology, especially when studying extremes. The most intense extreme precipitation events in France, particularly the Mediterranean episodes ("episodes"), occur during autumn (September to November). Using a standard calendar year (January to December) risks splitting a single autumn convective season or grouping two distinct autumn seasons into the same year, which violates the independence assumption of annual maxima. Starting the year on September 1 ensures that the entire autumn convective season is kept intact within a single year. We have added this explanation to the methodology section to justify our choice.

  Manuscript modifications: In `Section 2.3 (Methodology / Period definitions)` (formerly line 220), we have added the justification for the choice of the hydrological year:

  > *"The choice of the hydrological year instead of the calendar year is standard in French climatology and hydrology of extremes. In France, the most intense extreme precipitation events, particularly the Mediterranean episodes (e.g., Cévenol events), occur during autumn (September to November). Using a standard calendar year (January to December) would risk splitting a single autumn convective season across two different years, or grouping two distinct autumn seasons into the same year. This could violate the block-independence assumption required for the GEV modeling of annual maxima. Starting the hydrological year on September 1 ensures that each autumn convective season is kept fully intact within a single annual block."*

- **It can be confusing in the text with GEV models and Arome model, could consider naming one different.**

  Response: We agree. In the revised text, we refer to CNRM-AROME as the “climate model/simulation” and to the GEV models as “statistical models/distributions” to avoid any confusion.

  Manuscript modifications: Throughout the manuscript, we have revised the terminology to prevent confusion between the climate model and the extreme value models. Specifically:

  - We refer to CNRM-AROME as the “CP-RCM”, “climate model”, or “simulation”.
  - We refer to GEV models (M_0 to M_3^*) as “statistical models” or “statistical GEV distributions”.

- **To little font in Figure 8**

  Response: We have increased the font size of the labels and values in Figure 8 (and all other figures) to ensure they are fully legible in the final publication.

  Manuscript modifications: All labels and tick marks in Figure 8 (monthly GEV trend synthesis) have been enlarged to improve readability. Similar font size updates have been applied to Figures 4, 5, 6, and 7.